CH8.1 三角比

1.$\frac{5}{2}$　2.$\frac{41}{5}$　3.$\frac{1}{4}$　4.$\frac{7}{5}$　5.$\sqrt{37}$　6.2　7.$\frac{k}{\sqrt{1-k^2}}$ 8.$(2,-7)$ 9.$-\frac{7}{3}$ 10.$-\frac{3}{7}$

解析

1. $\sqrt{3}\tan 30° + \sqrt{2}\sin 45° + \cos 60°$

$$= \sqrt{3} \times \frac{1}{\sqrt{3}} + \sqrt{2} \times \frac{1}{\sqrt{2}} + \frac{1}{2} = \frac{5}{2}$$

2. $\frac{2\sin\theta+5\cos\theta}{4\sin\theta-\cos\theta} = \frac{2\times\frac{\sin\theta}{\cos\theta}+5}{4\times\frac{\sin\theta}{\cos\theta}-1} = \frac{2\tan\theta+5}{4\tan\theta-1}$

$$= \frac{2\times\frac{3}{7}+5}{4\times\frac{3}{7}-1} = \frac{6+35}{12-7} = \frac{41}{5}$$

3. $\sin 300°\tan 150° + \sin 150°\cos 120°$

$$= \sin(360°-60°)\tan(180°-30°) +$$

$$\sin(180°-30°)\cos(180°-60°)$$

$$= -\sin 60° \times (-\tan 30°) + \sin 30° \times (-\cos 60°)$$

$$= -\frac{\sqrt{3}}{2} \times \left(-\frac{1}{\sqrt{3}}\right) + \frac{1}{2} \times \left(-\frac{1}{2}\right) = \frac{1}{2} - \frac{1}{4} = \underline{\frac{1}{4}}$$

4. $\because \theta$ 為第二象限角

$$\therefore \sin\theta > 0, \cos\theta < 0 \Rightarrow \sin\theta - \cos\theta > 0$$

$$(\sin\theta + \cos\theta)^2 = \sin^2\theta + 2\sin\theta\cos\theta + \cos^2\theta = \frac{1}{25}$$

$$\Rightarrow \sin\theta\cos\theta = -\frac{12}{25}$$

又 $(\sin\theta - \cos\theta)^2 = (\sin\theta + \cos\theta)^2 - 4\sin\theta\cos\theta$

$$= \frac{1}{25} - 4 \times \left(-\frac{12}{25}\right) = \frac{49}{25}$$

$$\therefore \sin\theta - \cos\theta = \frac{7}{5}$$

5. 令 A,B 兩點之直角坐標為 $(x_1,y_1),(x_2,y_2)$

則 $x_1=4\cos300°=4\times\frac{1}{2}=2$

$$y_1=4\sin300°=4\times\left(-\frac{\sqrt{3}}{2}\right)=-2\sqrt{3}$$

$$x_2=3\cos60°=3\times\frac{1}{2}=\frac{3}{2}$$

$$y_2=3\sin60°=3\times\frac{\sqrt{3}}{2}=\frac{3\sqrt{3}}{2}$$

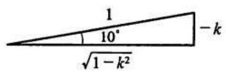
$$\overline{AB}=\sqrt{(x_1-x_2)^2+(y_1-y_2)^2}$$

$$=\sqrt{\left(2-\frac{3}{2}\right)^2+\left(-2\sqrt{3}-\frac{3\sqrt{3}}{2}\right)^2}=\sqrt{\frac{1}{4}+\frac{147}{4}}=\sqrt{37}$$

6. $\frac{\sin(90°+\theta)\cos(90°-\theta)}{\cos(180°-\theta)\sin(360°-\theta)}+\frac{\sin(180°-\theta)\cos(90°+\theta)}{\sin(180°+\theta)\cos(270°+\theta)}$

$$=\frac{\cos\theta\sin\theta}{(-\cos\theta)(-\sin\theta)}+\frac{\sin\theta(-\sin\theta)}{(-\sin\theta)(\sin\theta)}=1+1=\underline{2}$$

7. $\because\cos(-100°)=\cos(-90°-10°)=-\sin10°=k$



$\therefore\sin10°=-k$,如右圖

$$\tan890°=\tan(180°\times5-10°)$$

$$=-\tan10°=-\left(\frac{-k}{\sqrt{1-k^2}}\right)=\frac{k}{\sqrt{1-k^2}}$$

8. $f(\theta) = 4\cos^2\theta + 4\sin\theta - 3 = 4(1 - \sin^2\theta) + 4\sin\theta - 3$

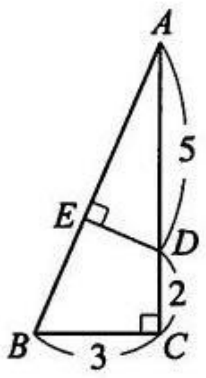
$$= -4\sin^2\theta + 4\sin\theta + 1 = -4\left(\sin\theta - \frac{1}{2}\right)^2 + 2$$

$$\because 0^\circ \le \theta < 360^\circ \ \therefore -1 \le \sin\theta \le 1$$

當 $\sin\theta = \frac{1}{2}$ 時， $f(\theta)$ 有最大值 2

當 $\sin\theta = -1$ 時， $f(\theta)$ 有最小值 -7

∴ 數對 $(M, m) = (2, -7)$



9. 如右圖 $\because \angle ADE = \angle B$

$$\therefore \tan\angle ADE = \tan\angle B = \frac{\overline{AC}}{\overline{BC}} = \frac{2+5}{3} = \frac{7}{3}$$

$$\tan\angle CDE = \tan(180^\circ - \angle ADE)$$

$$= -\tan\angle ADE = -\frac{7}{3}$$

10. $\overline{AE} = \overline{AB} - \overline{BE} = 10 - 7 = 3$

$\cos\theta = \cos(180^\circ - \angle AED) = -\cos\angle AED = -\frac{\overline{AE}}{\overline{DE}} = -\frac{3}{7}$